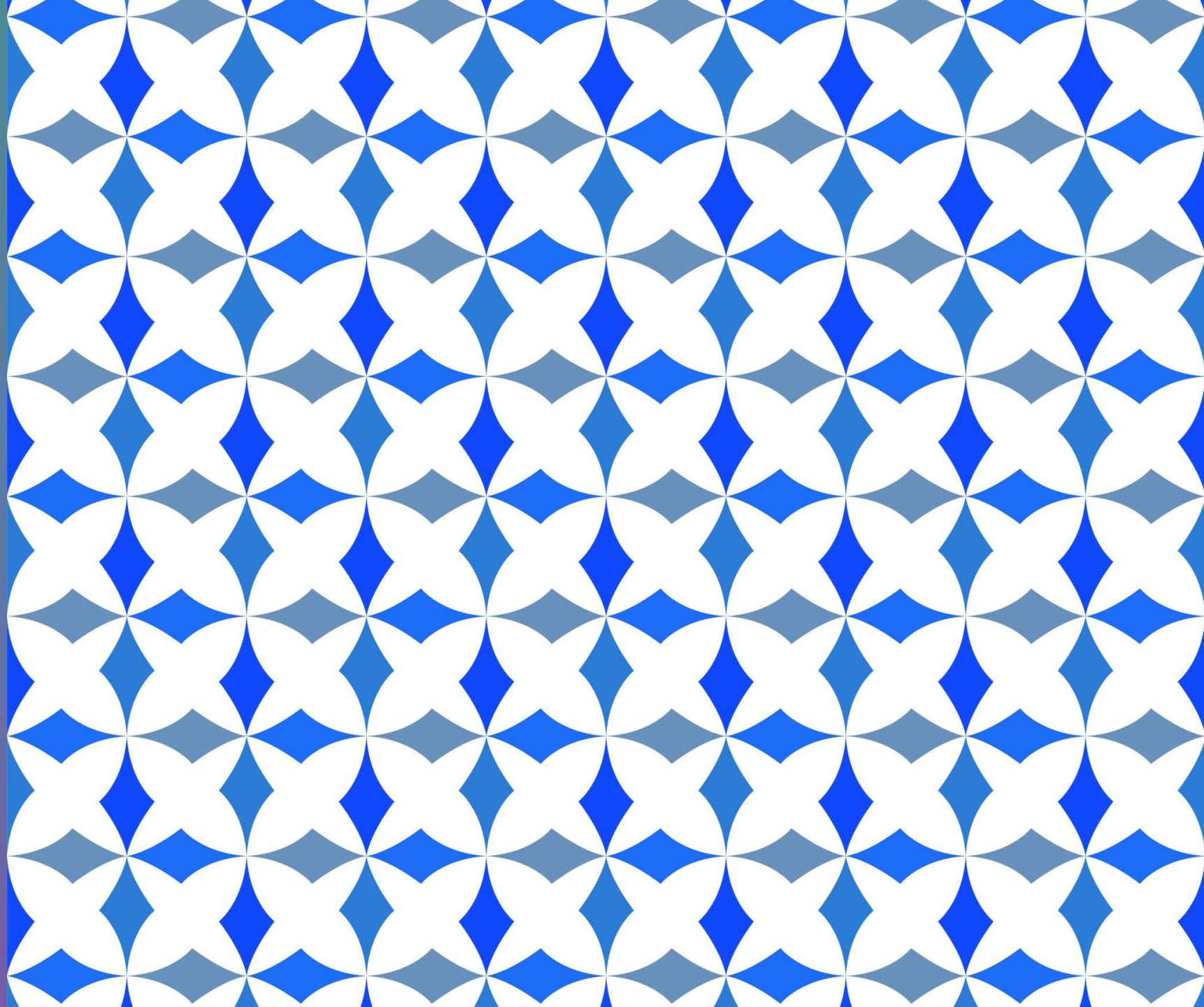


ENGCV370

DR MEGAN BOSTON

RISK,
HAZARD, AND
RESILIENCE
ASSESSMENTS



LEARNING OBJECTIVES

OUTLINE

Definitions

Hazard Identification
& Vulnerability
Assessment
(Wednesday)

Risk Analysis and
Tools (Thursday)

Finding Sources
(Friday)

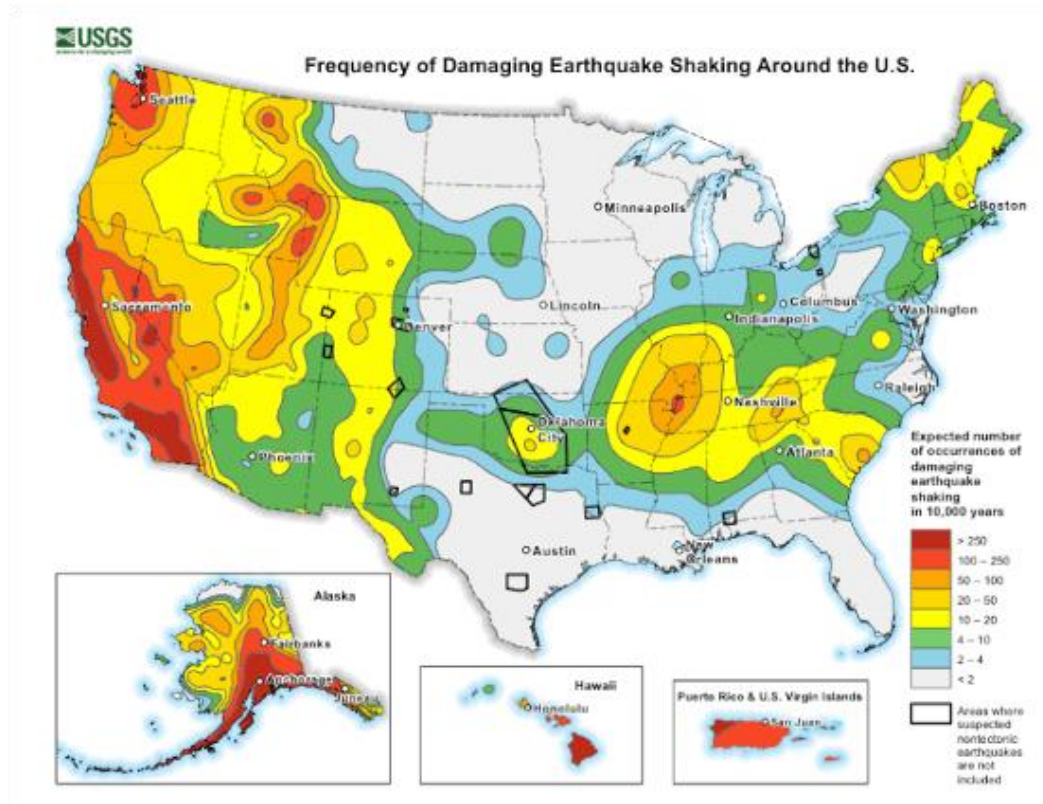
Resilience
(Wednesday)

Applying Resilience
(Thursday)

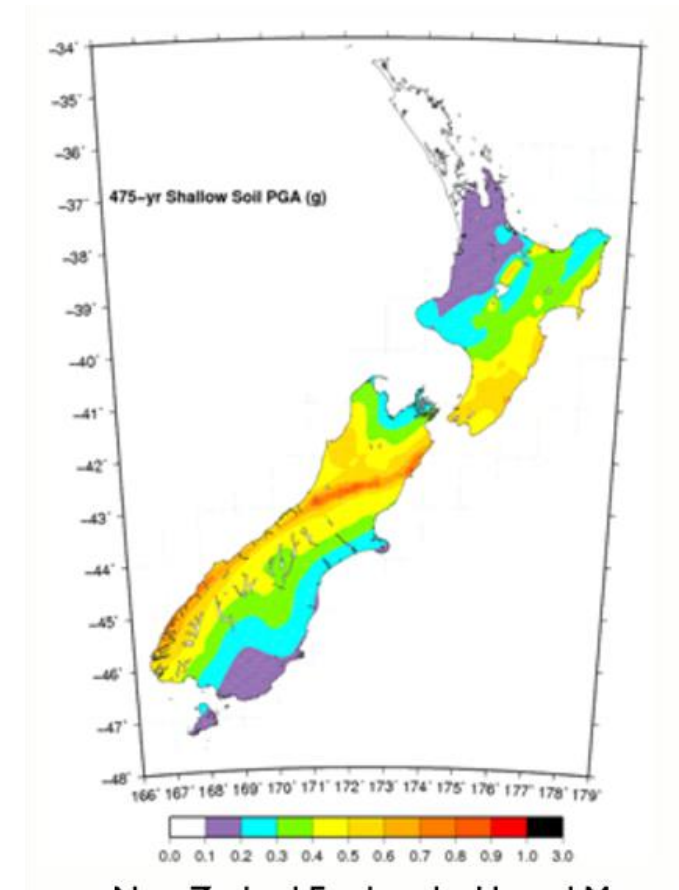
Resilience
Applications, Cultural
and Social (Friday)

Report help (Study
Break)

Hazard – A threat from a natural or manmade source that has the potential to cause property damage, loss of life, injury, social or economic disruption, or degrade the environment.



USGS Earthquake Hazard Map.
Source: USGS



New Zealand Earthquake Hazard Map.
Source: GNS

RISK

```
graph TD; RISK[RISK] --- Q1[What can go wrong? (scenarios combining hazards and outcomes)]; RISK --- Q2[How likely is it to go wrong? (probability of the scenario)]; RISK --- Q3[What are the consequences if it goes wrong?];
```

What can go wrong? (scenarios combining hazards and outcomes)

How likely is it to go wrong? (probability of the scenario)

What are the consequences if it goes wrong?

HAZARD

is the potential
to cause harm



when crossing a road,
cars are a **hazard**

RISK

is the likelihood of
harm taking place



when crossing
a highway, the risk of an
accident is **high**



when crossing
a country road, the risk of
an accident is **low**

HAZARD

in foods can be...



physical
for instance
pieces of bones
in fish products



biological
for instance
harmful bacteria,
viruses or parasites



chemical
for instance mercury
in fish or acrylamide
in starchy food

RISK

is determined by
the **exposure...**



how much

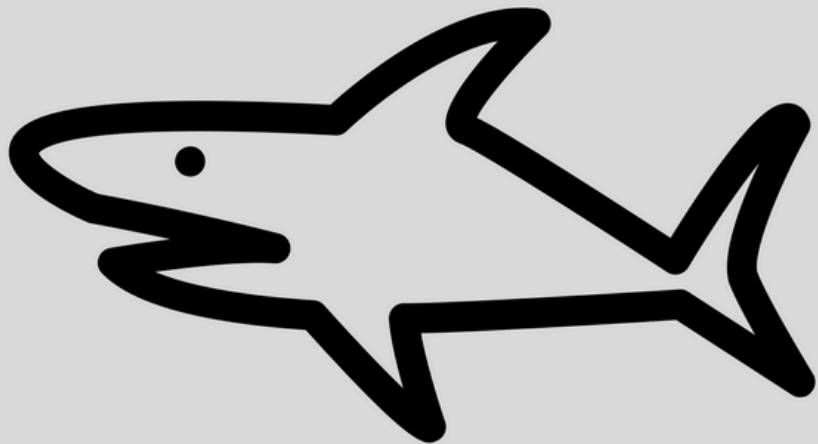
how long

how often

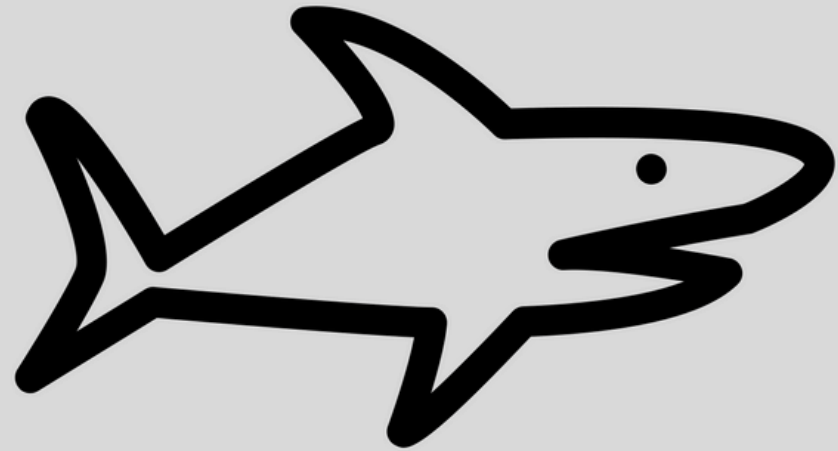
...to a hazard

Hazard

Something that can potentially cause harm



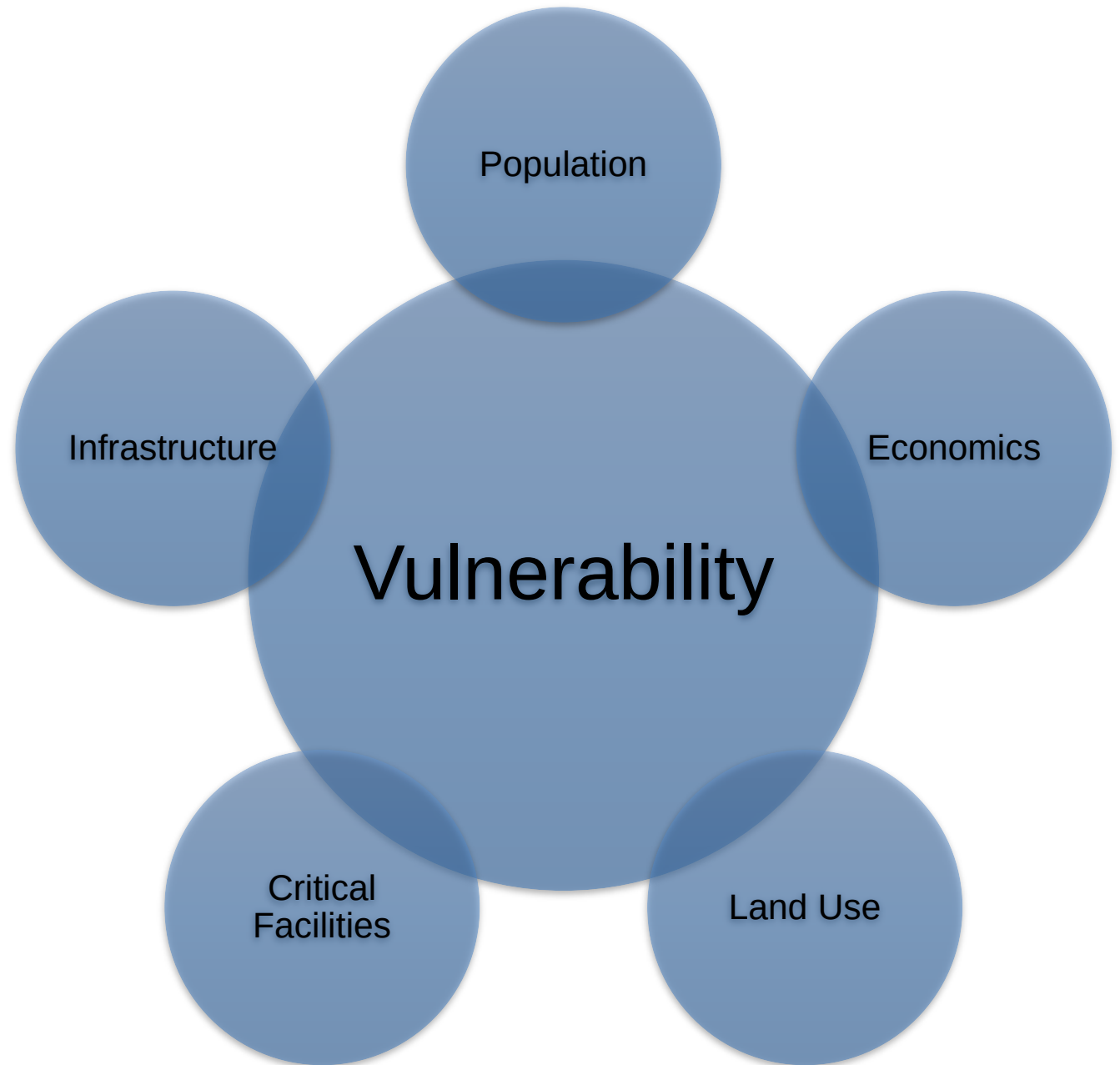
Risk



= hazard + exposure

VULNERABILITY

The diminished capacity of an individual or group to anticipate, cope with, resist and recover from the impact of a natural or man-made hazard



DISASTER

A serious disruption of the functioning of a community or a society at any scale due to hazardous events interacting with conditions of exposure, vulnerability and capacity, leading to one or more of the following: human, material, economic and environmental losses and impacts



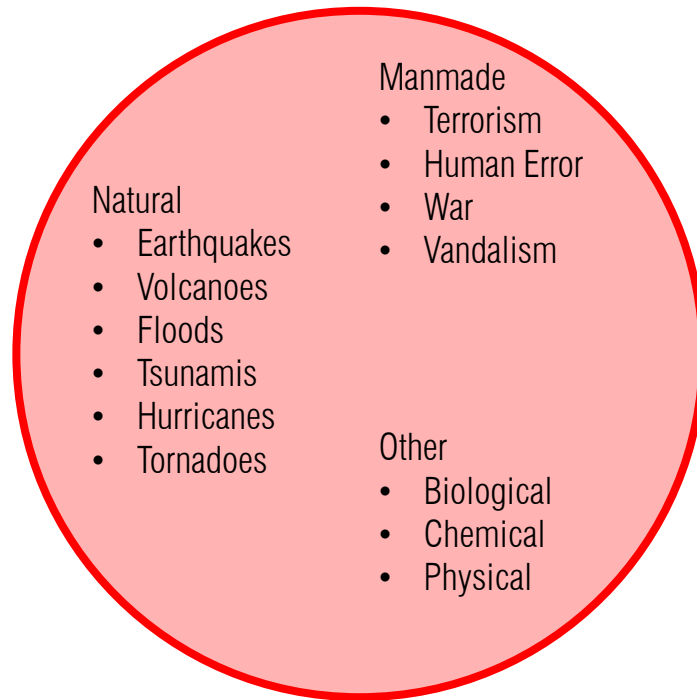
Downtown Christchurch after Feb 2011 Earthquake. Source: Wikimedia Commons

RESILIENCE

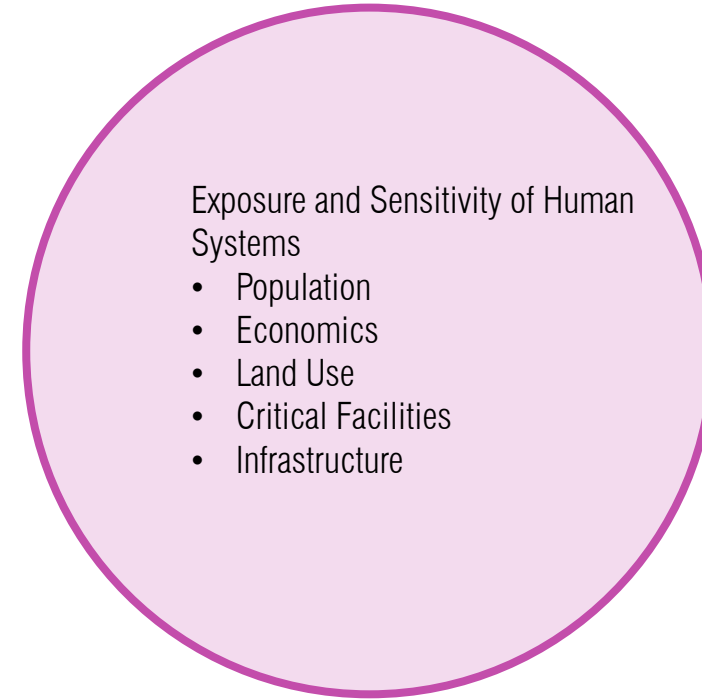
The capacity to absorb or mitigate the impact of hazard events while maintaining and restoring essential services

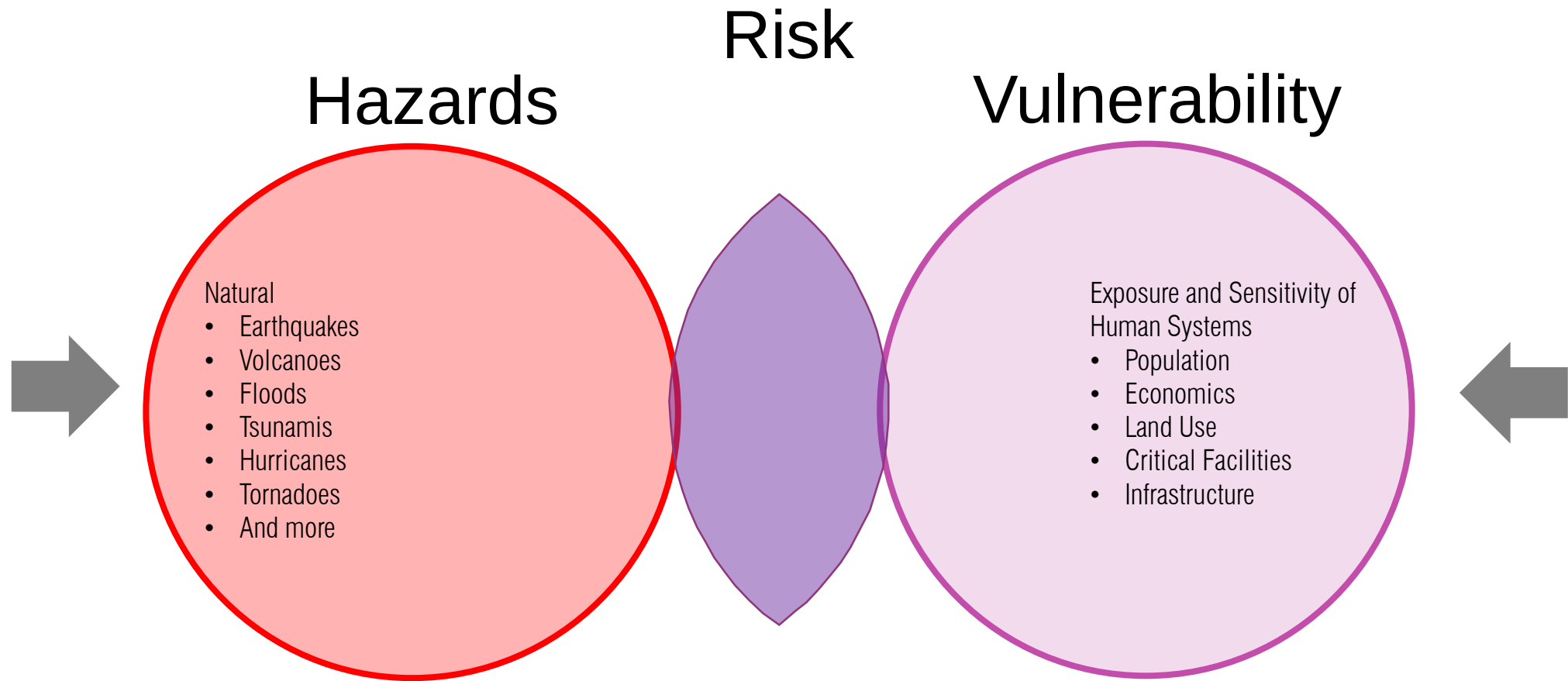


Hazards



Vulnerability





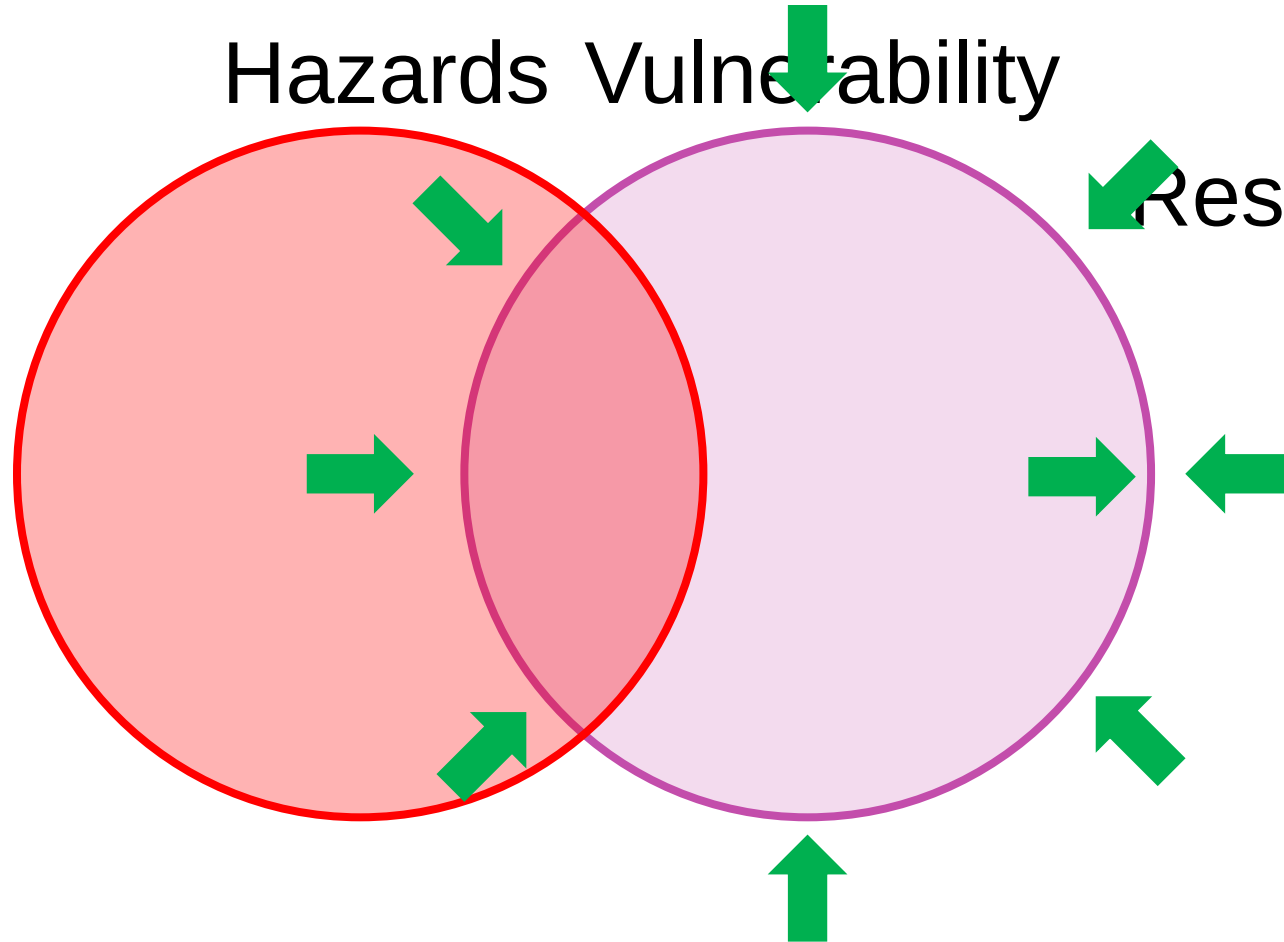


Volcanic eruption that is not near population centers or human infrastructure will have little to no impact on a community



Hazards Vulnerability

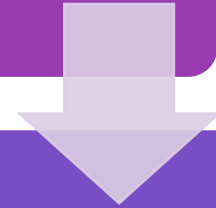
Resilience



RISK ASSESSMENT

HAZARD AND RISK ASSESSMENT

Methodical assessment of
the potential hazards



Determining potential
impacts from hazards



Determine the
vulnerabilities that are
making things susceptible
to hazards

HAZARD IDENTIFICATION

- What are the potential hazards for an engineering system?

TYPES OF HAZARDS

Natural

Anthropogenic

Technological

TYPES OF HAZARDS

UNDRR

Meteorological and Hydrological

Extraterrestrial

Geohazards

Environmental

Chemical

Biological

Technical

Societal

GH0003 / GEOHAZARDS / Seismogenic (Earthquakes)

Liquefaction (Earthquake Trigger)

Definition

Soil liquefaction occurs when soil is transformed from a solid to a liquid state as a result of increased pore pressure and reduced effective stress. It is typically caused by rapid loading of the soil during earthquake shaking (AGI, 2017).

Reference

AGI, 2017. Liquefaction [soil]. American Geosciences Institute (AGI). www.americangeosciences.org/word/liquefaction-soil Accessed 14 October 2002.

TL0019 / **TECHNOLOGICAL** / Cyber Hazard

Malware

Definition

Malware is a summary term for different forms of malevolent software designed to infiltrate and infect computers, typically without the knowledge of the owner (ITU, 2008).

Reference

ITU, 2008. ITU Study on the Financial Aspects of Network Security: Malware and Spam. International Telecommunication Union (ITU). www.itu.int/ITU-D/cyb/cybersecurity/docs/itu-study-financial-aspects-of-malware-and-spam.pdf Accessed 3 October 2020.

S00008 / **SOCIETAL / Economic**

Financial shock

Definition

A financial shock is an unexpected disturbance which originates from the financial sector and has a significant effect on an economy (e.g. national, regional, or global). The term is largely used to refer to events which have negative impacts (ECB, 2013).

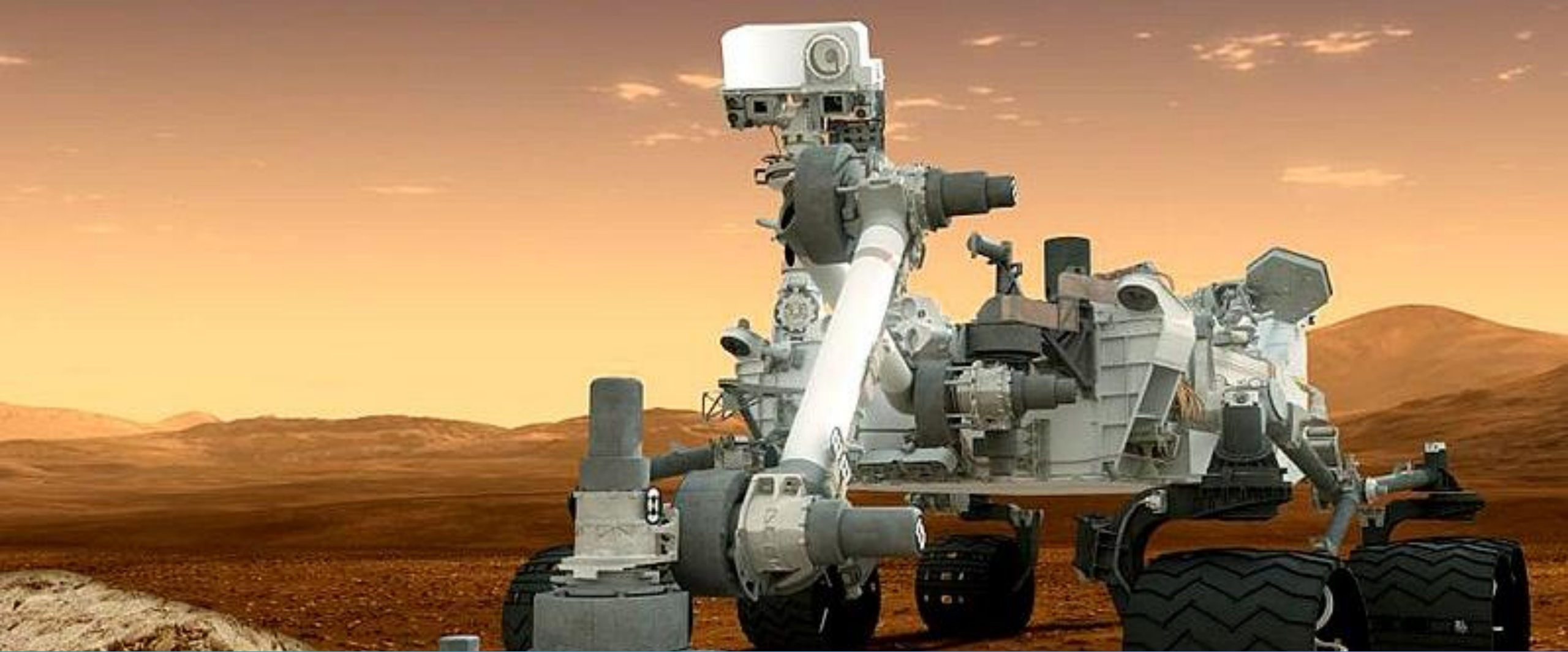
Reference

ECB, 2013. Financial Shocks and the Macroeconomy: Heterogeneity and Non-Linearities. European Central Bank (ECB), Occasional Paper Series No. 143. www.esm.europa.eu/sites/default/files/financialshocksandthemacroeconomy.pdf Accessed 12 September 2020.



HAZARD IDENTIFICATION: CASE STUDY

IMAGE: HAMILTON CITY COUNCIL

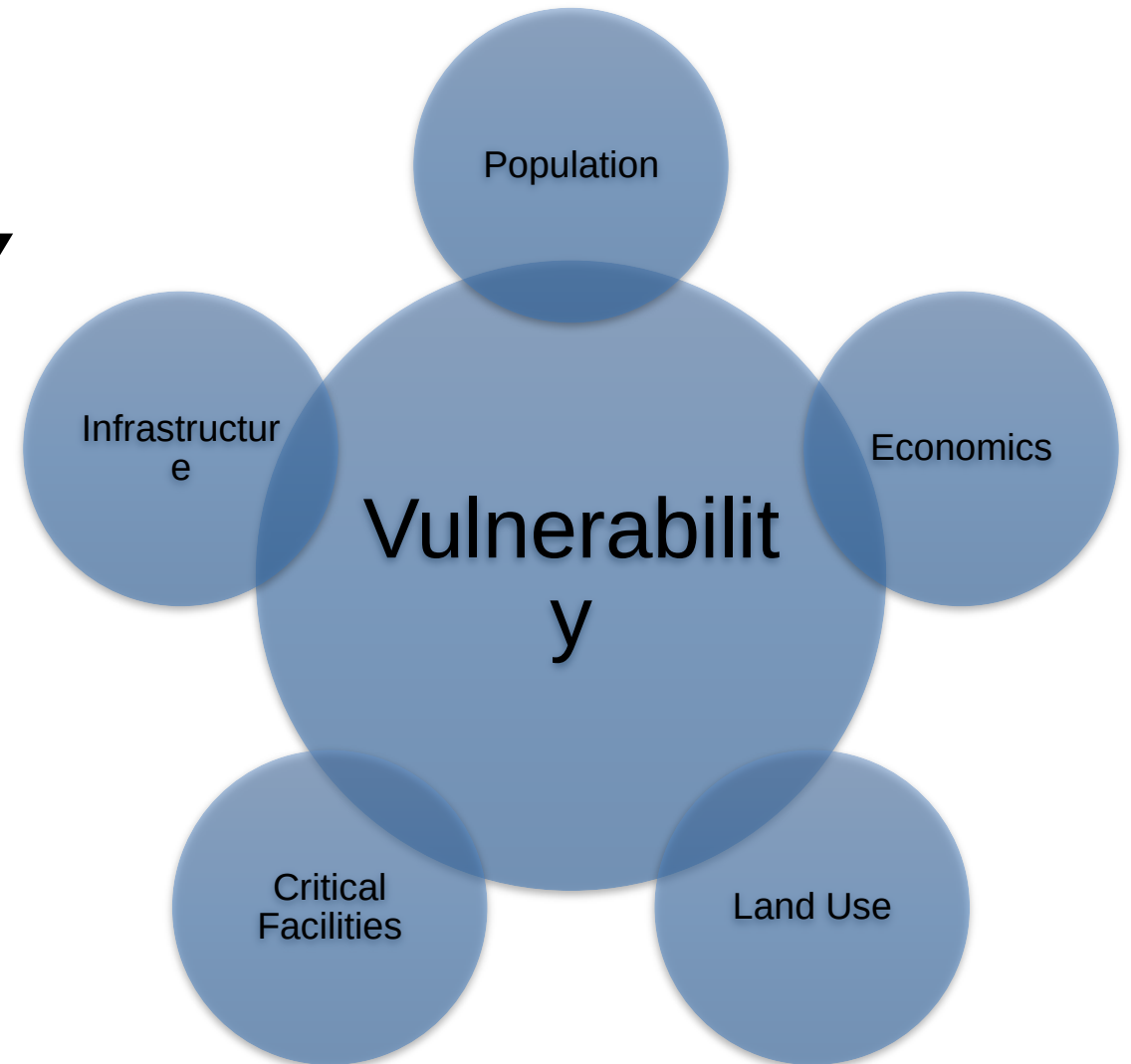


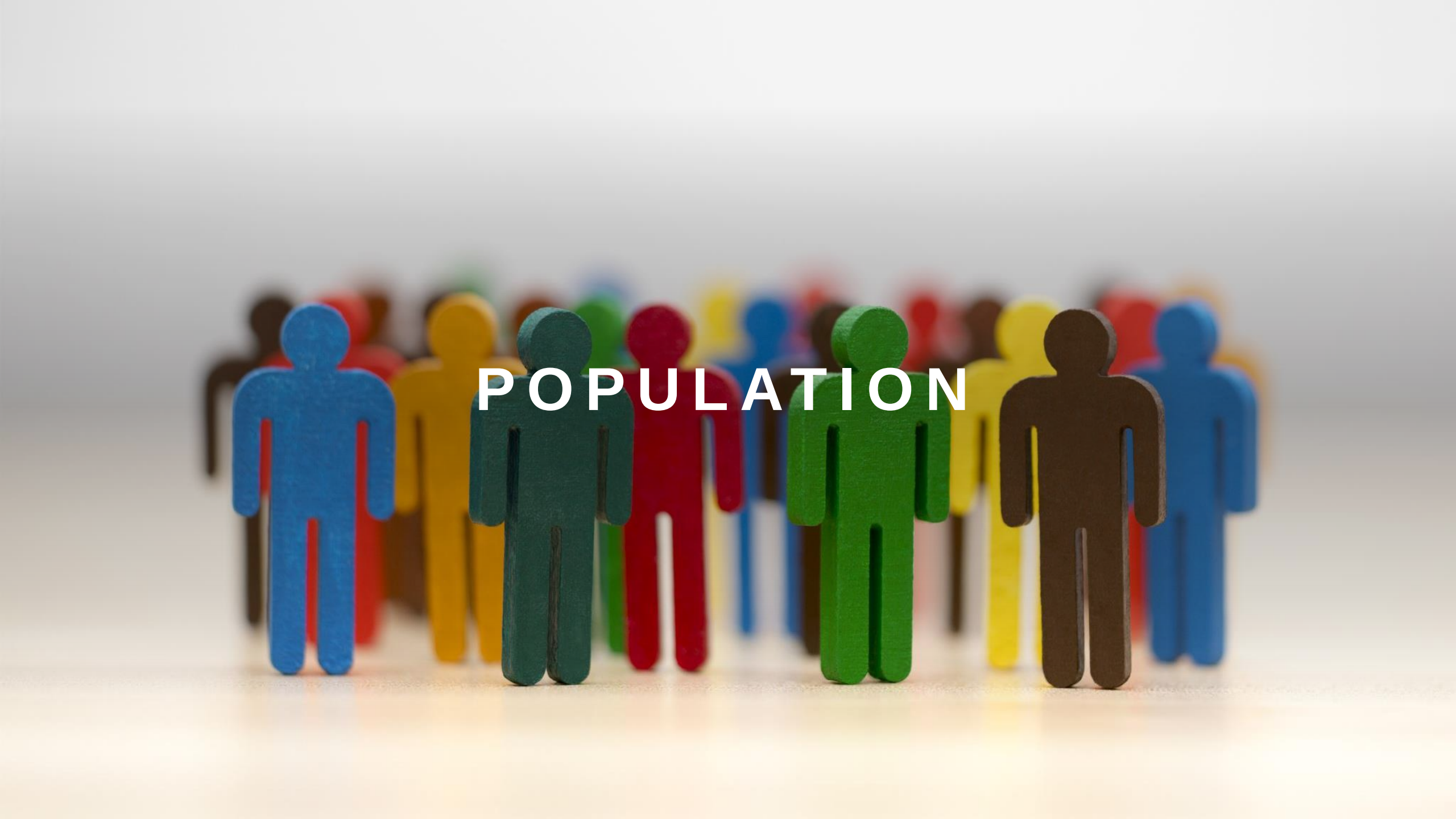
HAZARD IDENTIFICATION

VULNERABILITY ASSESSMENT

VULNERABILITY

- Identify all critical vulnerabilities
- Consider critical assets for the system



A group of colorful wooden human figures standing in a line, with the word 'POPULATION' overlaid in white text. The figures are in various colors including blue, yellow, red, green, and brown. The word 'POPULATION' is written in a bold, white, sans-serif font, centered over the figures. The background is a soft, out-of-focus light gray.

POPULATION



ECONOMICS

LAND USE





INFRASTRUCTURE